

# On Proof Systems for #QBF

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## Abstract

For a quantified Boolean formula (QBF), the problem of computing the number of winning strategies is known as the #QBF problem. This problem is considered harder than the analogous #SAT problem. Recently, important proof systems for QBFs and #SAT have been studied. By extending the ideas from both fields, we show that it is possible to design proof systems for #QBF. Such proof systems are important not only for advancing the theory of #QBF but also for certifying and designing better #QBF solvers, an area that is still in its early stages.

In this paper, we explore #QBF proof systems to count the number of Skolem functions. In addition to a naive system, we study #QBF systems based on the  $\forall$ -expansion rule of QBFs. We observe that these systems have inherent structural weaknesses that lead to lower bounds. As an alternative, we propose a #QBF proof system that we call Q-MICE, which consists of sound inference rules for computing and certifying the #QBF solution, similar to the line-based #SAT proof system MICE. To demonstrate the strength of Q-MICE, we present various upper bounds, such as the quantified version of the propositional XOR-PAIRS formula, which are known to be hard for MICE. Consequently, we also separate Q-MICE from  $\forall$ -expansion based #QBF proof systems.

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## 1 Introduction

#QBF, the problem of counting the number of winning strategies of a QBF, is challenging from several perspectives. Firstly, from a computational complexity point of view, #QBF is known to be #PSPACE-complete [13]. #PSPACE is at least as large as #P, but potentially much larger. Secondly, the number of winning strategies may be doubly exponential, beyond the mere exponential bounds of propositional model counting (#SAT). Several proof systems for the #SAT problem have been studied. Notable systems include KCPS(#SAT) [7] and CPOG [6] that use *knowledge-compilation*. These systems prove a model count by a transformation to a knowledge compilation class where model counting is easy. Recently, the CLIP system [9] worked using a connection between circuits and PSPACE. Another approach is to derive the model count via line-based systems, such as MICE (Model-counting Induction by Claim Extension [10]). Starting at axioms, it computes the model count using inference rules.

In this paper, by extending ideas from QBF and #SAT proof systems, we provide the



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first proof systems for the general #QBF problem, and study its proof complexity. Such proof systems are important to advance the theory of #QBF. The existence of proof systems can be useful towards #QBF solving, in both the design and correctness of the solvers.

The main contribution of the paper is extending the MICE proof system for #QBF (Section 3.2), denoted as Q-MICE. Q-MICE is a line based #QBF proof system, in which each line is of the form  $(Q.F, A, c)$ , where  $Q.F$  is a true QBF,  $A$  is a partial strategy that consists of Skolem functions for some existential variables, and  $c$  is the number of ways to extend  $A$  into complete winning strategies for the existential player of  $Q.F$ . Q-MICE consists of the axiom rule (Definition 6), the composition rule (Definition 8) and the join rule (Definition 11). Each rule can be applied if its corresponding conditions are satisfied, which are required for the rule to be sound. In addition to the soundness and completeness proof of Q-MICE (Theorem 14), we provide meaningful families with short proofs (Section 4) and establish an exponential separation between Q-MICE and the #QBF proof system  $\forall\text{Exp}+\text{MICE}$  (Theorem 15). The  $\forall\text{Exp}+\text{MICE}$  proof system (Section 3.1) semantically expands the  $\forall$  variables of the QBF (as in [12]), then finishes the proof with a #SAT proof system, in this case MICE.

**Related work:** Recently, some #QBF solvers [14, 16, 8, 15] have been introduced for solving the #QBF problem. For example, the d4-QBF [8] solver have been designed for #QBF which uses few techniques from SAT-solving and works on the full assignment tree of the QBF recursively along with a decomposition step ([8, Section 3.3]). The decomposition step identifies two or more non-connected components in the QBF matrix and separately computes the model-count in the components and merges them at the end. Another state-of-the-art #QBF solver for the restricted case, that is, for QBFs with two quantifier alternations, is the qCounter solver [14]. qCounter tries to enumerate winning strategies systematically one by one and then tries to certify that no more winning strategies exists.

## 2 Preliminaries

For a CNF  $\phi$  and its variables ( $\text{vars}(\phi)$ ), a partial assignment  $\alpha$  is a mapping from some variables ( $\text{vars}(\alpha) \subseteq \text{vars}(\phi)$ ) to  $\{0, 1\}$ .  $\phi|_\alpha$  denotes substituting values of  $\text{vars}(\alpha)$  from  $\alpha$  into  $\phi$ . If  $\phi|_\alpha$  is true,  $\alpha$  satisfies  $\phi$ . We can also extend the concept of restriction to functions which respect some total order of variables. E.g., if  $A := \{z = f(x, y), y = g(x)\}$ , and the ordering of variables is  $x \leq y \leq z$ ,  $\phi|_A$  denotes  $z$  substituted with  $f(x, y)$ , then  $y$  with  $g(x)$ .

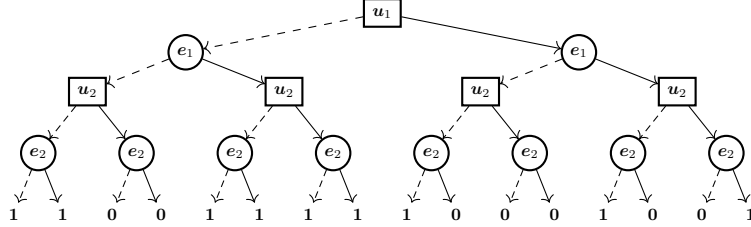
Quantified Boolean Formulas (QBFs) are an extension of propositional Boolean formulas where each variable is quantified with one of  $\{\exists$  (existential),  $\forall$  (universal) $\}$ . In this paper, QBFs (represented as  $Q.\phi$ ) are of the form  $Q_1X_1\dots Q_pX_p \cdot \phi(X_1 \cup \dots \cup X_p)$ , where  $X_i$ s are disjoint sets of variables;  $Q_i \in \{\exists, \forall\}$  and  $Q_i \neq Q_{i+1}$ ,  $Q$  is the quantifier prefix with  $p$  alternations and the matrix  $\phi$  is a CNF. The outermost (resp. innermost) quantified variables are  $X_1$  (resp.  $X_p$ ). If  $x \in X_i$ , for any  $y \in X_j$  where  $j < i$ , we say  $y$  occurs to the left of  $x$  in  $Q$  (i.e.  $y \leq_Q x$ ).  $L_Q(x)$  is the set of  $\forall$ -variables to the left of an  $\exists$ -variable  $x$ .

We can understand the semantics of QBFs as a game between a  $\forall$  and an  $\exists$  player, i.e., in the  $i^{\text{th}}$  step the player  $Q_i$  assigns Boolean values to the variables  $X_i$ . At the end, the  $\exists$  (resp.  $\forall$ ) player wins if substituting this complete assignment of variables in  $\phi$  evaluates to 1 (resp. 0). An  $\exists$  **strategy** is a set  $\mathcal{S}$  containing a Boolean function  $S_x$  for each  $x \in \text{vars}_\exists(Q)$  (called Skolem functions), where the decision for each  $x$  depends only on the variables in  $L_Q(x)$ , i.e. the Skolem function for  $x$  only takes  $|L_Q(x)|$ -many arguments, corresponding to the values of the universal variables in  $L_Q(x)$ , that would be played before  $x$ . Formally, for a true QBF  $Q.\phi$ , existential strategy  $\mathcal{S}$  is  $\{S_x(L_Q(x)) \mid x \in \text{vars}_\exists(Q)\}$ . The **winning strategy** for  $\exists$  is a strategy that for every possible assignment of  $\text{vars}_\forall(Q)$ , the Skolem functions of the

strategy respond on  $vars_{\exists}(Q)$  so that the QBF matrix is satisfied under the joint assignment to  $\exists$  and  $\forall$  variables. A QBF is true iff there exists a winning strategy for the  $\exists$ -player [2]. TQBF is the set of all true QBFs. There is a dual notion of falsity (FQBF) for winning strategies (sets of Herbrand functions) for the  $\forall$ -player, defined in a similar manner. Given a QBF, the problem of #QBF is to compute the number of winning strategies for its  $\exists$ -player.

A QBF  $Q.\phi$  can be represented as a full binary **assignment tree** where nodes in every level are labelled with one variable in the order of  $Q$  (i.e., the root is the outermost variable). Two outgoing edges from every node are labelled with 0 or 1. The leaves are labelled with 0/1, which is the value that  $\phi$  evaluates to when substituted with the complete assignment according to the edges on the root-to-leaf path. The satisfiability of the QBF is then computed in a bottom-up fashion (in the assignment tree) with the syntactic meaning of the variables. Here, we define subtree of the assignment tree as containing the root and at least one leaf. A **model-tree** (winning strategy) of a true QBF in this representation is a subtree with one outgoing edge for nodes with an  $\exists$ -variable and two outgoing edges for nodes with an  $\forall$ -variable, and all leaves of this tree being 1s. Given a true QBF, we can count the number of model-trees combinatorially with a bottom-up approach: values at the leaves are already 0/1, at the  $\exists$ -variables add the values of the children (because you only need one outgoing edge) and at the  $\forall$ -variables multiply the values of the children (because you need both outgoing edges). Finally at the root, the value equals the #QBF answer. For e.g., #QBF( $\Phi$ ) from Figure 1 is 4. For instance, an explicit winning strategy from the same Figure 1 is,  $\mathcal{S} = \{S_{e_1} = 1, S_{e_2} = u_2\}$ . We use the following running example throughout the paper.

► **Example 1.** Let  $\Phi := \forall u_1 \exists e_1 \forall u_2 \exists e_2 . (e_1 \vee \bar{u}_2) \wedge (\bar{u}_1 \vee u_2 \vee \bar{e}_2) \wedge (\bar{u}_1 \vee \bar{u}_2 \vee e_2)$ . The #QBF( $\Phi$ ) = 4, the complete assignment tree of  $\Phi$  is shown in Figure 1 for convenience.



■ **Figure 1** Assignment tree of the QBF  $\Phi$  from Example 1 (solid edge is 1 and dashed edge is 0)

*Verifying that two strategies are different:* Given two strategies  $\mathcal{S}_1, \mathcal{S}_2$  of  $\exists$ -player for a QBF  $Q.\phi$ , they are different if for some complete assignment  $\alpha$  to the  $\forall$  variables (witness assignment),  $\mathcal{S}_1, \mathcal{S}_2$  output different values for at least one of the Skolem functions corresponding to some  $\exists$ -variable  $x$ . That is, given two winning strategies  $\mathcal{S}_1 = \{S_{1,x}(L_Q(x)) \mid x \in vars_{\exists}(Q)\}$  and  $\mathcal{S}_2 = \{S_{2,x}(L_Q(x)) \mid x \in vars_{\exists}(Q)\}$ , they are different if there exists an  $x \in vars_{\exists}(Q)$  and a witness assignment  $\alpha$  to  $vars_{\forall}(Q)$  such that  $S_{1,x}(\alpha) \neq S_{2,x}(\alpha)$  (here,  $S_{1,x}(\alpha)$  ignores the values of the  $\forall$ -variables of  $\alpha$  which are not in  $L_Q(x)$ ). We opt for complete assignments for witnessing, as in general, the variable  $x$  is not known in advance). Given a witness assignment, it is easy to verify that the two strategies are different.

*Verifying correctness of a winning strategy:* In the QBF matrix, replacing all occurrences of  $\exists$ -variables with the Skolem functions of the given strategy leads to a propositional statement with only  $\forall$ -variables in it. A proof that this is a tautology certifies it is a winning strategy.

We use the Cook-Reckhow [1] notion of a proof system as a sound and complete polynomial

time checking function. For proof systems  $g_1, g_2$  for a language  $L$ ,  $g_1$  is exponentially separated from  $g_2$  if there is a family of  $x \in L$  for which there are small sized  $g_1$ -proofs but require exponential size  $g_2$ -proofs. For FQBFs (set of false QBFs), we can provide refutations by combining propositional inference with a rule for handling universal variables. ‘ $\forall$ red’ allows us to perform a 0 or 1 substitution on a universal variable  $u$ , appearing in a conjunct (i.e. any derived line). The condition is that no other variable quantified to the right of  $u$  is in that particular conjunct. In the refutations we provide in this paper we will use Frege+ $\forall$ red [4] refutations. Frege systems consist of a complete set of axioms schemas that define the Boolean connectives ( $\perp, \top, \neg, \rightarrow, \leftrightarrow, \wedge, \vee$ ), and a Modus ponens rule. Frege is chosen here because it is a powerful proof system. We will omit individual propositional inference steps.

### 3 Proof Systems for #QBF

Naively, we can define a rudimentary proof system for the #QBF problem. The system enumerates all  $k$  winning strategies for the given true QBF  $Q.\phi$ . Then, to argue the correctness of the count, the system also proves: 1) the correctness of each winning strategy, which requires a tautological proof of the propositional statement obtained after replacing  $\exists$ -variables with their respective Skolem functions in  $Q.\phi$ . 2) that the strategies are all distinct from each other, with explicit witness assignments for distinguishing each pair of winning strategies. 3) that no other winning strategy exists for  $Q.\phi$  which we encode succinctly (in the size of  $k$  and  $Q.\phi$ ) as a false QBF  $\Psi$  as follows.

▷ **Claim 2.** Given a true QBF  $Q.\phi$  and a set of winning strategies  $\mathcal{S}_1, \dots, \mathcal{S}_k$ , the following QBF is false if and only if there are no other winning strategies for  $Q.\phi$ :

$$\Psi := \exists \vec{\alpha}_1, \dots, \vec{\alpha}_k \ Q.\phi \wedge \bigwedge_{i \in [k]} \left( \bigwedge_{u \in U} (u \leftrightarrow \alpha_{i,u}) \rightarrow \bigvee_{y \in E} (y \not\leftrightarrow S_{i,y}(\vec{\alpha}_i)) \right)$$

where,  $U = \text{vars}_{\forall}(Q)$ ,  $E = \text{vars}_{\exists}(Q)$  and  $\vec{\alpha}_i$  is a witness assignment of the fact that  $\mathcal{S}_i$  differs from any potential new winning strategy for  $\Psi$ , and  $\alpha_{i,u}$  is the Boolean assignment corresponding to universal variable  $u$  in  $\alpha_i$ . That is  $\alpha_i = \{\alpha_{i,u} \mid u \in U\}$ .  $S_{i,y}$  is the Skolem function of the existential variable  $y$  in the winning strategy  $\mathcal{S}_i$ .

*Proof.* Consider any winning strategy  $\mathcal{S}$  for the  $\exists$ -player of  $\Psi$ , which would contain an assignment for  $\vec{\alpha}_1, \dots, \vec{\alpha}_k$ . Now, for  $\Psi$  to be true,  $\mathcal{S}$  must differ from each  $\mathcal{S}_i$  with  $\vec{\alpha}_i$  as a witness. Moreover, the Skolem functions in  $\mathcal{S}$  for each  $x \in \text{vars}_{\exists}(Q)$  must also satisfy  $\phi$  on each assignment to  $\text{vars}_{\forall}(Q)$ . Hence, any winning strategy for  $\Psi$  corresponds to a  $k + 1^{\text{th}}$  distinct winning strategy for  $Q.\phi$ , and vice-versa. Therefore,  $\Phi$  is a false QBF if and only if there are no other winning strategies for  $Q.\phi$ . ◁

Observe that, as opposed to the qCounter solver [14], our naive system is capable of certifying the correct #QBF number of winning strategies and with no restrictions on the number of alternations on the given QBF. However, any such system has trivial lower bounds whenever the number of distinct winning strategies is (doubly) exponential.

#### 3.1 An Expansion-based #QBF Proof System: $\forall\text{Exp}+\text{MICE}$

We can avoid an enumerative approach to build advanced proof systems. The expansion-based approach [12] for QBF solving is a well studied technique which can be applied for #QBF. We have to transform the #QBF problem to a #SAT problem and preserve the model count. To be precise, given a QBF  $Q.\phi$ , one can expand it over all universal variable

assignments so that the resulting formula consists of clauses over only existential variables. To keep track of the existential variable dependency to the universal variable, we annotate the existential literal  $e$  with a universal variable assignment over  $L_Q(e)$  making a new existential variable from  $e$  [12]. For example, consider a QBF  $\exists e_1 \forall u_1 \exists e_2 \forall u_2 \exists e_3. (e_1 \vee \bar{u}_2 \vee e_2 \vee u_1 \vee e_3)$ . Let  $\alpha = \{u_1 = 0, u_2 = 1\}$  be the assignment to all universal variables of QBF, then the corresponding axiom clause in the expanded CNF is  $(e_1 \vee e_2^{u_1/0} \vee e_3^{u_1/0, u_2/1})$ . This is sound because of the following.

► **Proposition 3.** *A true QBF  $\Phi$ , when expanded with all complete assignments to universal variables, leads to a CNF formula  $\phi$  such that every model of  $\phi$  leads to a distinct model of  $\Phi$  and vice versa. In other words, their model-counts are the same.*

In  $\forall\text{Exp}+\text{MICE}$ , one needs to expand the QBF with all possible assignments to universal variables. Otherwise, if one expands on a smaller set of assignments to obtain a CNF  $\phi' \subset \phi$ , one can only guarantee  $\#\text{QBF}(\Phi) \leq \#\text{SAT}(\phi')$ . This is because, for any CNF  $\phi'$  obtained after dropping a few clauses from a CNF  $\phi$ , we have  $\#\text{SAT}(\phi') \geq \#\text{SAT}(\phi)$ .

► **Corollary 4.** *For any  $\#\text{SAT}$  proof system  $\mathcal{P}$ , the proofs in the  $\#\text{QBF}$  proof system  $\forall\text{Exp}+\mathcal{P}$  for a QBF  $\Phi$  need exponential size in terms of  $|\text{vars}_{\forall}(\Phi)|$ .*

## 3.2 A Line-based $\#\text{QBF}$ Proof System: Q-MICE

In  $\forall\text{Exp}+\text{MICE}$ , eliminating all universal quantifiers at the beginning of the proof can cause an exponential explosion, even where the model count may otherwise be straightforward. In this section, we introduce a line-based  $\#\text{QBF}$  proof system Q-MICE. The idea is that interleaving quantifier handling and other rules allows better control of the proof size in instances where the model count ought to be easy to determine. Q-MICE is inspired from MICE for  $\#\text{SAT}$  [10]. A claim in MICE is of the form  $(F, A, c)$  where  $F$  is a CNF formula,  $A$  is a partial assignment, and  $c$  is the correct number of satisfying assignments to  $F|_A$ . Below, we define the analogous Q-MICE system for  $\#\text{QBF}$ , using similar notations from [5] for ease.

### 3.2.1 Inference rules for Q-MICE

In Q-MICE, we use the claims of the following format:  $(Q.F, A, c)$ , where  $Q$  is the quantifier prefix.  $F$  is a CNF formula (matrix),  $c$  is a non-negative integer written in decimal<sup>1</sup> and  $A$  is a partial strategy (i.e., some of the  $\exists$ -variables  $x$  are allotted strategies in the form of a Boolean circuit computing a Skolem function  $S_x(L_Q(x))$ ). The set of these variables is represented by  $\text{vars}(A)$ . So  $A$  is a collection of circuits, one for each  $x \in \text{vars}(A)$ . A claim  $(Q.F, A, c)$  is said to be a **valid claim** if  $c$  is the correct number of ways to extend  $A$  (by assigning strategies to the remaining  $\exists$ -variables, say  $A'$ ) such that  $A \cup A'$  is a winning strategy for the  $\exists$ -player in  $Q.F$ . The **size** of a claim is the number of symbols needed to represent the claim in some fixed representation. For example, circuits in  $A$  can be represented as CNFs using Tseitin transformation [17]. We need the following definitions for Q-MICE.

► **Definition 5** (Restriction of  $\forall$ -variables in a QBF). *Let  $Q.F$  be a QBF. By a restriction of the  $\forall$ -variable  $(y = b \in \{0, 1\})$  in a QBF  $Q.F$  (denoted as  $Q.F|_{\{y=b\}} = Q'.F \wedge (y = b) = Q'.F'$ ), we mean that the quantification of  $y$  is flipped to  $\exists$  in the prefix  $Q'$  and an additional unit clause forcing the restriction  $y = b$  (i.e. if  $b = 1$  then  $(y)$  else  $(\bar{y})$ ) is added in the matrix. All other clauses in  $F$  remain.*

<sup>1</sup> Future improvements could use succinct integer representations to avoid trivial exponential lower bounds.

- **Definition 6** (Axiom rule). For a QBF  $Q.F$ , one can derive:  $\overline{(Q.F|_\rho, A, c)}$  where
- (A-1)  $\rho$  is a (partial) assignment to  $\text{vars}_\forall(Q)$  and  $Q.F|_\rho$  is denoted as  $Q'.F'$  (as in Def 5).
  - (A-2)  $F|_{\{\rho \cup A\}}$  is a tautology and an explicit proof of this is provided.
  - (A-3)  $\text{vars}(A) \subseteq \text{vars}_\exists(Q)$ , calculate  $c$  as follows:
    - Initially,  $\ell = 1$ . Starting from the innermost quantified variable in  $Q$  (say  $x$ ):
    - if  $x \in \text{vars}_\exists(Q)$  and  $x \notin \text{vars}(A)$  then:  $\ell \leftarrow 2 \cdot \ell$  (i.e. double the count),
    - if  $x \in \text{vars}_\forall(Q)$  and  $x \notin \text{vars}(\rho)$  then:  $\ell \leftarrow \ell^2$  (i.e. square the count),
    - at the end of the reverse quantification sequence,  $c \leftarrow \ell$ .

The intuition of Axiom rule is as follows: let  $\phi$  be a tautology, for any quantification  $Q$  to  $\text{vars}(\phi)$ ,  $Q.\phi$  will be a true QBF. In fact, all strategies are winning in  $Q.\phi$ , so model-counting is easy. For instance, in the QBF from Example 1, we can derive the following axioms:  $(\Phi|_{\{u_1=0\}}, \{e_1 = 1\}, 4)$ ,  $(\Phi|_{\{u_2=0\}}, \{e_1 = \bar{u}_1, e_2 = 0\}, 1)$ . Tautology proofs here are trivial.

► **Lemma 7.** Any claim  $(Q.F|_\rho, A, c)$  derived by using the Axiom rule of Definition 6 is valid i.e.,  $c$  is the correct number of models for the QBF  $Q.F|_\rho \wedge \bigwedge_{x \in \text{vars}(A)} (x \leftrightarrow A(x))$ .

**Proof.** In this rule, we find a subtree  $(Q'.F' \cup A)$  of the assignment tree  $(Q.F)$  which has all 1-leaves and compute all possible valid models in this subtree. This computation is just combinatorial (described in Section 2). For the correctness of this model-count, we argue inductively on the subtree  $(Q'.F' \cup A)$ : Base case: the model-count at the leaves is 1. Inductively, model-count at unrestricted existential nodes  $x$  i.e.  $x \notin \text{vars}(A)$  (or universal nodes  $x$  i.e.  $x \notin \text{vars}(\rho)$ ) is the addition (or multiplication) of model-counts at their two children owing to the structure of any model-tree. The remaining two cases of induction are restricted existential variables  $x$  (i.e.  $x \in \text{vars}(A)$ ) or universal variables  $x$  (i.e.  $x \in \text{vars}(\rho)$ ) of  $Q$ . The model-count in these cases is retained from the only remaining child as both are now existentially quantified in  $Q'$  and need the only remaining outgoing edge in any model-tree. ◀

► **Definition 8** (Composition rules). Given a QBF  $Q.F$ ,

1. (Composition-a:) 
$$\frac{(Q.F|_\rho, A_1, 1), \dots, (Q.F|_\rho, A_n, 1)}{(Q.F|_\rho, A, n)}, \text{ where}$$
  - (C-a1) for all  $i \in [n]$ ,  $\text{vars}(A_i) = \text{vars}_\exists(Q)$  and  $\text{vars}(\rho) \subseteq \text{vars}_\forall(Q)$ .
  - (C-a2) for  $i \neq j \in [n]$ ,  $A_i \neq A_j$  (providing witness assignment where  $A_i, A_j$  disagree).
  - (C-a3)  $A \subseteq \bigcap_{i \in [n]} A_i$  (a subset of strategies where all  $A_i$ s are syntactically equal)
2. (Composition-b:) 
$$\frac{(Q.F|_\rho, A_1, c_1)}{(Q.F|_\rho, A, c_1)}, \text{ where}$$
  - (C-b1)  $A \subseteq A_1 \subseteq \text{vars}_\exists(Q)$  and  $\text{vars}(\rho) \subseteq \text{vars}_\forall(Q)$ .
3. (Composition-c:) 
$$\frac{(Q.F|_\rho, A_1, c_1), \dots, (Q.F|_\rho, A_n, c_n)}{(Q.F|_\rho, A, \sum_{i \in [n]} c_i)}, \text{ where}$$
  - (C-c1)  $\text{vars}(A_1) = \dots = \text{vars}(A_n) \subseteq \text{vars}_\exists(Q)$  and  $\text{vars}(\rho) \subseteq \text{vars}_\forall(Q)$ .
  - (C-c2) for  $i \neq j \in [n]$ ,  $A_i \neq A_j$  and  $A \subseteq \bigcap_{i \in [n]} A_i$
  - (C-c3) Let  $x$  be the innermost  $\exists$ -variable in  $Q$  that  $\in A_1 \setminus A$ .
    - All  $y \in \text{vars}_\exists(Q)$  and  $y \leq_Q x \implies y \in \text{vars}(A_i)$ .
    - All  $y \in \text{vars}_\forall(Q)$  and  $y \leq_Q x \implies y \in \text{vars}(\rho)$ .

For ease, we denote  $Q.F|_\rho$  as  $Q'.F'$  in these rules. All these rules additionally need an FQBF-proof of the **absence of models statement**:  $\Psi := Q'.F' \cup A \cup \{\bar{A}_i \setminus \bar{A}\}_{i \in [n]}$  where  $n$  is the number of hypothesis claims.  $\Psi$  encodes as a QBF the negation of the fact that there exists no other winning strategies for the  $\exists$ -variables in  $\text{vars}(A_i) \setminus \text{vars}(A)$ . The exact encoding of this statement is provided in the next claim.

▷ **Claim 9.**  $\Psi$  in the composition rules (Definition 8) can be encoded as a short QBF.

**Proof.** This encoding is a generalization of that defined in Claim 2. For an input QBF  $Q.F$ , the composition rule in general is of the form where there are  $n$  partial strategies  $A_1, \dots, A_n$  over the same set of existential variables  $X \subseteq \text{vars}_{\exists}(Q)$  in the hypothesis, and a subset of these strategies ( $X' \in X$ ) which appear in all of them are retained as  $A$  in the conclusion. The ‘absence of models’ statement encodes that there are no other winning strategies (in  $Q'.F'$ ) for the variables in  $X \setminus X'$  when also adhering to strategies from  $A$ .

Recall that  $A_i$ s are a set consisting of one function  $S_{A_i, x}$  for every  $x \in \text{vars}(A_i)$ . So, in  $Q'.F'$ , enforcing the strategies in  $A$  is simply adding clauses  $(x \leftrightarrow S_{A, x})$  for every  $x \in X'$ .

Now to see if there is any  $n + 1^{\text{th}}$  winning strategy for variables  $\in X \setminus X'$ , we need it to be different than those already in the hypotheses. As discussed in Section 2, for two strategies to be considered different, there must be some witness assignment ( $\alpha$  below) to  $\forall$ -variables of  $Q'$  (say  $U'$ ) such that the value of the function computed by these strategies is different. Now, if the following QBF encoding is true, it implies that there indeed is a  $n + 1^{\text{th}}$  strategy which is missing in the hypothesis and the rule cannot be used.

$$\Psi := \exists \vec{\alpha}_1, \dots, \vec{\alpha}_n \ Q'.F' \wedge \bigwedge_{x \in X'} (x \leftrightarrow S_{A, x}) \wedge \bigwedge_{i \in [n]} \left( \bigwedge_{u \in U'} (u \leftrightarrow \alpha_{i, u}) \rightarrow \bigvee_{y \in X \setminus X'} (y \not\leftrightarrow S_{A_i, y}(\vec{\alpha}_i)) \right)$$

However, if the above encoding is a false QBF, it certifies that there is no such missed strategy and we can proceed to drop the strategy restrictions on variables  $\in X \setminus X'$ . ◁

The intuition for the composition rules: Composition-a is useful to directly count complete winning strategies. Composition-b is useful when some variables have unique winning Skolem functions in the subtree. Composition-c is useful for combining subtrees which all agree on the restrictions of universal variables and individually are a path graph from the root to the innermost variable where the hypotheses strategies differ. E.g., in Example 1, we can use composition-b to derive:  $\frac{(\Phi|_{u_1=1}, \{e_1=1, e_2=u_2\}, 1)}{(\Phi|_{u_1=1}, \{e_1=1\}, 1)}$ ; The absence of models statement  $\Psi$  is:  $\exists \alpha_1 \exists u_1 \exists e_1 \forall u_2 \exists e_2 . (e_1 \vee \bar{u}_2) \vee (\bar{u}_1 \vee u_2 \vee \bar{e}_2) \wedge (\bar{u}_1 \vee \bar{u}_2 \vee e_2) \wedge (u_1) \wedge (e_1) \wedge ((u_2 = \alpha_1) \rightarrow (e_2 \neq \alpha_1))$ . For an FQBF-proof of  $\Psi$  in Frege+ $\forall$ red [4], we use propositional inferences to derive  $(u_2 \vee \alpha_1)$  and  $(\bar{u}_2 \vee \bar{\alpha}_1)$ . Using universal reduction to drop variable  $u_2$ , we can derive  $(\alpha_1)$  and  $(\bar{\alpha}_1)$ , which gives the needed contradiction. Finally, we use composition-a (or -c) to derive:  $\frac{(\Phi|_{u_1=0, u_2=0}, \{e_1=0, e_2=0\}, 1) , (\Phi|_{u_1=0, u_2=0}, \{e_1=0, e_2=1\}, 1)}{(\Phi|_{u_1=0, u_2=0}, \{e_1=0\}, 2)}$ . Here, the corresponding absence of models statement is trivially false.

► **Lemma 10.** Any claim  $(Q'.F', A, c')$  derived by using the composition rules of Definition 8 from valid hypothesis claims is also valid.

**Proof.** In composition-a, every hypothesis claim considers a full strategy for the existential player. That is, each hypothesis is a distinct model for the subtree corresponding to  $Q'.F' \cup A$  and no other models exist for the same (due to absence of models proof). Therefore, adding them up is the correct number of models for the formula  $Q'.F' \cup A$ .

In composition-b, there is only one hypothesis strategy for variables in  $\text{vars}(A_1) \setminus \text{vars}(A)$  and no other winning strategies are possible due to the absence of models proof. So, there is only one useful branch from these  $\exists$ -variables in the subtree of the assignment tree, and these branches need to be in any model tree. Therefore, the count at the root remains the same.

In composition-c,  $x$  is the innermost variable from  $\text{vars}(A_i) \setminus \text{vars}(A)$  and every variable to the left of  $x$  either has some strategy in every  $A_i$  or is restricted with 0/1 in the QBF. Hence, the corresponding subtree of the assignment tree for these hypotheses is a line graph

from root to  $x$ . Let  $y$  be the outermost variable from  $\text{vars}(A_i) \setminus \text{vars}(A)$ . In the subtree corresponding to  $Q'.F' \cup A$ , it is a line graph upto  $y$  and it only branches on every  $\exists$ -variable  $\in \text{vars}(A_i) \setminus \text{vars}(A)$  upto  $x$  and at  $x$  it has one branch with count  $c_i$  and the other with count 0 (due to the absence of models proof). Now the total count at  $y$ -node is just combinatorial and equal to  $\sum_{i \in n} c_i$ . As it is just a line graph till the root, this value propagates as is. ◀

► **Definition 11** (Join rule). For a QBF  $Q.F$ ,

$$\frac{(Q.F|_{\rho \cup \{y=0\}}, A, c_1), (Q.F|_{\rho \cup \{y=1\}}, A, c_2)}{(Q.F|_{\rho}, A, c_1 \cdot c_2)}, \text{ when}$$

(J-1)  $\rho$  should be (partial) assignment to  $\text{vars}_{\forall}(Q)$  and  $\text{vars}(A) \subseteq \text{vars}_{\exists}(Q)$ .

For ease, quantifier in the hypothesis is  $Q'$  and  $Q.F|_{\rho}$  is denoted as  $Q''.F_3$ .

(J-2)  $y \in \text{vars}_{\forall}(Q)$ ,  $y \in \text{vars}_{\exists}(Q')$ ,  $y \in \text{vars}_{\forall}(Q'')$  and unit clauses  $(y)$ ,  $(\bar{y})$  are dropped.

(J-3) All  $x \in \text{vars}_{\exists}(Q)$  and  $x \leq_Q y \implies x \in \text{vars}(A)$ .

(J-4) All  $x \in \text{vars}_{\forall}(Q)$  and  $x \leq_Q y \implies x \in \text{vars}(\rho)$ .

The intuition behind the join rule is to combine two subtrees (say  $G_1, G_2$ ) agreeing everywhere except for one restriction of a  $\forall$ -variable ( $y$ ) and  $G_1, G_2$  are simple path graph from the root to  $y$ . In Example 1, using join we can derive: 
$$\frac{(\Phi|_{u_1=0}, \{e_1=1\}, 4), (\Phi|_{u_1=1}, \{e_1=1\}, 1)}{(\Phi, \{e_1=1\}, 4)}$$

► **Lemma 12.** Any claim  $(Q.F|_{\rho}, A, c_1 \cdot c_2)$  derived by using the join rule of Definition 11 from valid hypothesis claims is also valid.

**Proof.** In the hypothesis,  $y$  is a  $\forall$ -variable of  $Q.F$  which is restricted and is now a  $\exists$ -variable in  $Q'$ . Every variable to the left of  $y$  either has some strategy in  $A$  or is restricted with 0/1 in  $\rho$  (due to J-3, J-4). Hence, the corresponding subtree of the assignment tree for these hypotheses is a path graph from root to  $y$ . The counts at root of hypotheses are the same at the  $y$ -node. Now, the conclusion subtree  $(Q''.F_3 \cup A)$  is similarly a path graph from root up to  $y$  and at  $y$  branches with 0, 1. Now, any model of this subtree should have both branches in its model-tree, therefore  $c_1 \cdot c_2$  is the correct number of models in the subtree. ◀

We now define the #QBF proof system Q-MICE using the inference rules introduced above.

► **Definition 13** (Q-MICE). A Q-MICE proof of a QBF  $Q.F$  with  $k$  winning strategies for the existential player is a sequence of lines  $L_1, \dots, L_s$  where  $L_s$  contains the claim  $(Q.F, \emptyset, k)$  and every line  $L_i$  for  $i \in [s]$  is derived by one of the following rules:

1. Axiom rule from Definition 6 along with the required tautology proof.
2. One of the Composition rules from Definition 8 with hypothesis from  $L_1, \dots, L_{i-1}$ , along with witnesses separating the hypothesis (partial) strategies, and absence of models proof
3. Join rule from Definition 11 with hypothesis from  $L_1, \dots, L_{i-1}$ .

The ‘length’ of a Q-MICE proof is the number of lines in it, and its ‘size’ is the size of all claims along with additional witnesses, and FQBF-proofs and tautology proof sizes.

► **Theorem 14.** Q-MICE is a polynomial-time verifiable sound and complete #QBF system.

**Proof.** Since all the inference rules are sound (Lemmas 7,8,12), the Q-MICE proof system is sound. For completeness: Use Axiom rule for every winning strategy of  $Q.F$  with  $c = 1$ . Using Composition-a rule with  $A = \emptyset$ , add these axiom claims to derive the correct #QBF answer: If you consider all winning strategies in the absence of models statement, it is indeed a false QBF and any complete FQBF-proof system suffices. Also every rule of Q-MICE is easily verifiable, hence any Q-MICE-proof is verifiable in time polynomial in the proof-size. ◀

Note that if the input QBF is false, one can use any one of the composition rules with no hypothesis strategies to derive the answer that #QBF answer is 0. However, this is equivalent to directly using a FQBF proof system to prove that the input QBF is false.

## 4 Upper bounds

In this section, we show the strength of Q-MICE by explicitly providing upper bounds for two example QBF families that have roots in theory [5] and practice [11], respectively.

### 4.1 A QBF family based on XOR-PAIRS

$$\mathcal{Q}.\oplus_n := \forall u_1, \dots, u_n \exists_{i \neq j \in [n]} x_{i,j} \cdot \bigwedge_{i \neq j \in [n]} (x_{i,j} \leftrightarrow u_i \oplus u_j)$$

$\mathcal{Q}.\oplus_n$  has  $n^2$  variables out of which  $n$  are universal variables, hence it requires super-polynomial ( $\Omega(2^n)$ ) sized proofs in the  $\forall\text{Exp}+\text{MICE}$  system (Corollary 4).

Below, we show that  $\mathcal{Q}.\oplus_n$  has a constant length and polynomial size Q-MICE proof  $\pi = \{L_1, L_2\}$ , where  $L_1$  is derived using the Axiom rule consisting of the entire QBF  $\mathcal{Q}.\oplus_n$ , a complete winning strategy  $A = \{S_{x_{i,j}} = u_i \oplus u_j \mid i \neq j \in [n]\}$  and the count 1, accompanied by a tautology proof of  $\oplus_n|_A$ . Line  $L_2$  is derived by applying the Composition-b rule on  $L_1$  by dropping the strategy restrictions in  $A$  accompanied by a FQBF proof of the absence of models statement. This maintains the previous count, hence  $\#\text{QBF}(\mathcal{Q}.\oplus_n) = 1$ . Precisely:

(Axiom rule)  $L_1 := (\mathcal{Q}.\oplus_n, A := \{S_{x_{i,j}} = u_i \oplus u_j \mid i \neq j \in [n]\}, 1)$ ; the tautology proof of  $\oplus_n|_A$  is a straightforward case of any introduction rule of  $\leftrightarrow$  in any Frege system.

(Composition-b rule)  $L_2 := (\mathcal{Q}.\oplus_n, \emptyset, 1)$ ; the absence of models statement as per Claim 9 is:

$$\begin{aligned} \Psi := & \exists \vec{\alpha}_1 \forall u_1, \dots, u_n \exists_{i \neq j \in [n]} x_{i,j} \cdot \bigwedge_{i \neq j \in [n]} (x_{i,j} \leftrightarrow u_i \oplus u_j) \\ & \wedge ((u_1 = \alpha_{1,u_1} \wedge \dots \wedge u_n = \alpha_{1,u_n}) \rightarrow (\bigvee_{i \neq j \in [n]} x_{i,j} \neq \alpha_{1,u_i} \oplus \alpha_{1,u_j})) \end{aligned}$$

FQBF proof of  $\Psi$  using Frege+ $\forall\text{red}$  : (for simplicity, we denote  $\alpha_{1,u_i}$  as  $\alpha_i$ )

$$\begin{array}{c} x_{i,j} \leftrightarrow u_i \oplus u_j \quad (u_1 = \alpha_1 \wedge \dots \wedge u_n = \alpha_n) \rightarrow (\bigvee_{i \neq j \in [n]} x_{i,j} \neq \alpha_i \oplus \alpha_j) \\ \hline (u_1 = \alpha_1 \wedge \dots \wedge u_n = \alpha_n) \rightarrow (\bigvee_{i \neq j \in [n]} u_i \oplus u_j \neq \alpha_i \oplus \alpha_j) \quad (\text{Prop. Frege inference}) \\ \hline (\alpha_1 = \alpha_1 \wedge \dots \wedge \alpha_n = \alpha_n) \rightarrow (\bigvee_{i \neq j \in [n]} \alpha_i \oplus \alpha_j \neq \alpha_i \oplus \alpha_j) \quad (\forall\text{red with } u_i = \alpha_i \text{ for } i \in [n]) \\ \hline \perp \quad (\text{Prop. Frege inference}) \end{array}$$

### 4.2 An Indexed Affine QBF Family

$$\Gamma = \mathcal{Q}.\chi_n := \forall u_1 \dots u_{\log n} \exists x \forall v_1 \dots v_n \exists y_1 \dots y_n \cdot \bigwedge_{i \neq (u)} (y_i = v_i \oplus x) \wedge (y_u = v_u \oplus \bar{x})$$

The QBF  $\Gamma$  can be thought of as a simple encryption process, where the function for  $x$  w.r.t. the index ' $u$ ' is the encryption scheme, which is applied to the plain-text ' $v$ ' to obtain ' $y$ ' as the cipher-text. Since  $\Gamma$  does not have a single variable  $u$ , representing  $y_u, v_u$  is challenging for a CNF encoding. However,  $u$  has only  $\log n$  bits so one way to represent  $y_u, v_u$  is to go through all the assignment space of  $\{0, 1\}^{\log n}$  in some order. Following this idea, the CNF encoding of  $\chi_n$  would take  $O(n^2)$  clauses of the following type (for all indexes  $\{0\}^{\log n}$  to  $\{1\}^{\log n}$ , where  $\{a\}^b$  is ' $a$ ' repeated  $b$  times,  $2^{\log n} = n$  such cases):

$$(\langle u_1, \dots, u_{\log n} \rangle = \{0\}^{\log n} \rightarrow (v_0 = y_0 \oplus \bar{x}) \wedge \bigwedge_{i \in [n] \setminus 0} (v_i = y_i \oplus x))$$

Based on the values of  $u = \langle u_1, \dots, u_{\log n} \rangle$ ,  $x$  and all  $v_i$  for  $i \in [n]$ , there is a single way to pick the value of all  $y_i$ s for  $i \in [n]$  to make  $\Gamma$  true and the number of possible Skolem functions for  $x$  (i.e.  $S_x(u_1, \dots, u_{\log n})$ ) is  $2^n$ . Since  $\Gamma$  has  $n + \log n$   $\forall$ -variables and  $2^n$  winning strategies, it requires exponential-size proofs in both the naive system and  $\forall\text{Exp}+\text{MICE}$  (Corollary 4).

A linear length and polynomial size Q-MICE proof for  $\Gamma$  is as follows (for ease, we represent  $\langle u_1 \dots u_{\log n} \rangle$  as  $\vec{u}$ ): For each  $\beta \in [n]$ , derive the following (below  $i \in [n] \setminus \beta$ ):

$$\begin{array}{c}
\frac{}{(\Gamma|_{\vec{u}=\beta}, \{x=0, y_\beta = \overline{v_\beta}, y_i = v_i\}, 1)} \text{ (Axiom)} \quad \frac{}{(\Gamma|_{\vec{u}=\beta}, \{x=1, y_\beta = v_\beta, y_i = \overline{v_i}\}, 1)} \text{ (Axiom)} \\
\frac{}{(\Gamma|_{\vec{u}=\beta}, \{x=0\}, 1)} \text{ (Comp-c)} \quad \frac{}{(\Gamma|_{\vec{u}=\beta}, \{x=1\}, 1)} \text{ (Comp-b)} \\
\hline
L'_\beta := (\Gamma|_{\vec{u}=\beta}, \emptyset, 2)
\end{array}$$

We used composition-b and composition-c rules in the above derivation. These rules require an FQBF proof of the corresponding absence of models statements. We next define the absence of model statement for the composition-b rules. For ease, we represent  $\langle v_1 \dots v_n \rangle$  as  $\vec{v}$  below. For each  $\beta \in [n]$  and  $q \in \{0, 1\}$ , we give the absence of models statement  $\Psi_{\beta, x=q}$  for the Composition-b rule below (for ease, we only show useful clauses in the matrix):

$\Psi_{\beta, x=q} := \exists \vec{\alpha}_1 \exists \vec{u} \exists x \forall \vec{v} \exists y_1, \dots, y_n \cdot (\vec{u} = \beta \rightarrow (v_\beta = y_\beta \oplus \bar{x}) \wedge \bigwedge_{i \in [n] \setminus \beta} (v_i = y_i \oplus x))$   
 $\wedge (x = q) \wedge (\vec{u} = \beta) \wedge ((\vec{v} = \vec{\alpha}_1) \rightarrow (y_\beta \neq \alpha_{1, v_\beta} \oplus \bar{x}) \vee \bigvee_{i \in [n] \setminus \beta} (y_i \neq \alpha_{1, v_i} \oplus x))$   
 FQBF proof of  $\Psi_{\beta, x=q}$  in Frege+ $\forall$ red proof system (after propagating  $x = q$  and  $\vec{u} = \beta$ ) is given below. For simplicity we denote  $\alpha_{1, v_i}$  as  $\alpha_i$  and the soundness of inference-lines  $A, B, C$  are explained after the derivation:

$$\begin{array}{c}
\frac{(v_\beta = y_\beta \oplus \bar{q}) \wedge \bigwedge_{i \in [n] \setminus \beta} (v_i = y_i \oplus q) \quad \left( (\vec{v} = \vec{\alpha}_1) \rightarrow (y_\beta \neq \alpha_\beta \oplus \bar{q}) \vee \bigvee_{i \in [n] \setminus \beta} (y_i \neq \alpha_i \oplus q) \right)}{\left( (\vec{v} = \vec{\alpha}_1) \rightarrow (v_\beta \oplus \bar{q} \neq \alpha_\beta \oplus \bar{q}) \vee \bigvee_{i \in [n] \setminus \beta} (v_i \oplus q \neq \alpha_i \oplus q) \right)} \text{ (A)} \\
\hline
\left( (\vec{\alpha}_1 = \vec{\alpha}_1) \rightarrow (\alpha_\beta \oplus \bar{q} \neq \alpha_\beta \oplus \bar{q}) \vee \bigvee_{i \in [n] \setminus \beta} (\alpha_i \oplus q \neq \alpha_i \oplus q) \right) \text{ (B)} \\
\hline
\perp \text{ (C)}
\end{array}$$

In this FQBF proof, we are only left with the soundness of inference lines A, B and C. (A): The hypothesis is the input QBF and the conclusion is derived by propositional Frege inferences which substitute  $y_i, y_\beta$  variables with the functions  $v_i \oplus q, v_\beta \oplus \bar{q}$  respectively. (B): Since  $\vec{v}$  are the rightmost variables in the hypothesis clause, the conclusion is derived by a  $\forall$ red rule which substitutes  $\vec{v} = \vec{\alpha}_1$ . (C): The hypothesis is a contradiction with true  $\rightarrow$  false.

For each  $\beta \in [n]$ , the absence of models statement for the Composition-c rule is :

$\Psi_\beta := \exists \vec{\alpha}_1, \vec{\alpha}_2 \exists \vec{u} \exists x \forall \vec{v} \exists y_1, \dots, y_n \cdot \chi_n \wedge (\vec{u} = \beta) \wedge ((\vec{v} = \vec{\alpha}_1) \rightarrow (x \neq 0)) \wedge ((\vec{v} = \vec{\alpha}_2) \rightarrow (x \neq 1))$ . Since  $\vec{v} \geq_{\Psi_\beta} x$ , the Frege+ $\forall$ red proof of  $\Psi_\beta$  is easy: use  $\forall$ red rule to substitute  $\vec{v} = \vec{\alpha}_1$  and  $\vec{v} = \vec{\alpha}_2$  in the two axiom clauses to derive  $(x \neq 0)$  and  $(x \neq 1)$ , a contradiction. So far in Q-MICE proof, we have derived lines  $L'_\beta$  for every  $\beta \in [n]$ . Now we can apply the join rule  $n - 1$  times and remove the restriction of universal variables  $u_{\log n}$  to  $u_1$  (in this order). This results in the last claim being  $(\Gamma, \emptyset, 2^n)$ .

► **Theorem 15.** *The #QBF proof system Q-MICE is exponentially separated from the naive #QBF system and the  $\forall$ Exp+MICE proof system.*

## 5 Discussion and Future Work

The paper proposes a #QBF proof system Q-MICE based on the #SAT proof system MICE [10] and proves that it is exponentially stronger than the naive (Section 3) and  $\forall$ Exp+MICE proof systems (Section 3.1). In fact we observe that Q-MICE is capable of certifying all the rules of d4-QBF [8] except the decomposable rule. However, Q-MICE is modular, in the sense that, based on the requirements, one can always add more sound #QBF inference rules like the decomposable rule from [8]. The challenge of doubly exponential solutions may be mitigated in Q-MICE by representing integers by arithmetic circuits. An immediate open problem is to establish a genuine lower bound [3] for Q-MICE. We conjecture that the propositional XOR-PAIRS (which are hard for MICE [5]) would be hard for Q-MICE but it is not a genuine lower bound. Another open problem is to extend other #SAT proof systems for #QBF.

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